

Rhea Madhogarhia

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Education

The University of Chicago | Chicago, IL | Graduating June 2026

- **B.S. in Computer Science** (Specialization: Machine Learning) & **B.A. in Cognitive Science**

Honors Thesis | In progress, expected Spring 2026 | Advisors: Jason Bridges, Monica Rosenberg

- *Working Title: Causal and Intentional Architectures of Mind: An Exploration of the Human-AI Relationship*

Autumn 2025 Coursework

- Machine Learning, Machine Learning for Computer Systems, Philosophy of AI, Truth (Linguistics)

Research Experience

Research Assistant | Cognition, Attention, and Brain (CAB) Lab | Chicago, IL

May 2024 – Present

PI: Prof. Monica Rosenberg

- Co-authored a preprint on neural reinstatement during narrative comprehension (contributions: conceptualization, data curation, investigation, writing/editing).
- Developing a methodological comparison of NLP techniques (LLMs, BERT, Flair) for sentiment analysis on transcribed fMRI speech data.
- Executing Representational Similarity Analysis (RSA) and sentiment scoring to map neural alignment with linguistic features in fMRI data, data processing, coding, and ML-assisted transcription of fMRI audio data.

Publications

Academic Publication (Preprint)

- Song, H., Ke, J., **Madhogarhia, R.**, Leong, Y. C., & Rosenberg, M. D. (2025). *Cortical reinstatement of causally related events sparks narrative insights by updating neural representation patterns*. bioRxiv. doi: [10.1101/2025.03.12.642853](https://doi.org/10.1101/2025.03.12.642853)

Industry Experience

Machine Learning Engineering Intern | Reddit | New York, NY

June 2025 – August 2025

- Engineered and productionized a novel, signal-driven First Pass Ranker (FPR) in Go and Python to improve search relevance using historical user engagement ("Clicks" and "Good Visits").
- Designed and tuned a ratio-driven boosting algorithm using Bayesian Optimization to prioritize high-value content and penalize clickbait.
- Authored formal design docs and experiment specifications, managing the full project lifecycle from implementation to A/B testing and production shipment.

Data Science & Machine Learning Intern | Charactour | Chicago, IL

June 2024 – August 2024

- Performed exploratory data analysis on personality traits from 9M+ users using Python (Pandas).
- Utilized BERT Regression to discover correlations between media properties and their fanbases.

Machine Learning & Data Intern | Blue Fever | Los Angeles, CA

June 2023 – August 2023

- Drove development of an automated user-reply program (prompt engineering, Python, and SQL/BigQuery).
- Crafted replies by leveraging experience as a help-line supporter, prioritizing authenticity and sensitivity.

- Conducted sentiment analysis on user data and presented key product insights to C-Suite leadership.

Conferences & Summits

- Madhogarhia, R. (2025, October). Attendee. *Grace Hopper Celebration (GHC)*, Chicago, IL.
- Madhogarhia, R. (2025, September). Accepted Participant. *CTRL + Future: Youth AI Forum*, Virtual.
- Madhogarhia, R. (2024). Invited Attendee. *The Rithm Project: Human Connection + AI Summit*, Lone Rock, CO.

Professional Affiliations & Service

Young Futures Foundation | *Research Analyst & Proposal Reviewer* *April 2024 – Present*

- Served on review committees for three funding cohorts and developed scorecards to evaluate grant proposals focused on youth wellbeing and responsible technology.

The Rithm Project | *Inaugural Cohort Member* *April 2024 – Present*

- Invited member of a professional network dedicated to discussing human connection and responsible technology.

Off-Off Campus | *Trainer, Improviser, Writer & Director* *September 2022 – Present*

- Cast in 37th Generation, committed to daily 3-hour rehearsals, practicing active listening and collaborative storytelling. Made best friends. Became an improviser, actor, and comedic writer. Performed long-form improv in New York Comedy Clubs during a Spring Break tour.
- Co-director, co-writer, stage designer, and actor in “Homage to Cable” @ The Second City

WHPK 88.5fm | *Public Affairs Show Creator, DJ, & Co-Host ("Finders, Keepers")* *December 2023 – Present*

- Curate and host a weekly musical interview show; conduct meaningful, personal interviews with authors, musicians, and community DJs for the South Side of Chicago radio audience.

Voices In Your Head | *A Cappella Vocalist, Sound System Manager, Co-Choreographer* *September 2023 – Present*

- Alto, Tenor, and Baritone in an all-gender a cappella group that competed in the 2025 ICCA Finals

Honors & Awards

- **Quad Faculty Research Grant** | University of Chicago (2024–2025)
- **People's Choice Award Winner** | UC Crew Team Digital Archive, UChicago Summer Tech Showcase 2024
- **Jeff Metcalf Fellowship** | University of Chicago (2023)

Technical Skills

- **Programming Languages:** Python, Java, Go, C, SQL, JavaScript, HTML/CSS, Bash
- **ML & Data Science:** Natural Language Processing (NLP), Bayesian Optimization, Model Finetuning, Experiment Design, Pandas, NLTK, BERT, Scikit-learn, HuggingFace
- **Developer Tools & Platforms:** Git, GitHub, Google BigQuery, Google Colab, MySQL, Vim
- **Development Practices:** Technical Documentation (Design Docs), A/B Testing, Agile (Scrums, Stand-ups)